



CS5071NI - Professional and Ethical Issues

100% Individual Coursework

2025-26 Spring

Credit: 15 Semester Long Module

Student Name: DEWANSHI SHARMA

London Met ID: 24046374

College ID: :NP0NT4A240219

Assignment Due Date: Monday, April 27, 2026

Assignment Submission Date: Monday, April 27, 2026

Word Count: 2721

I confirm that I understand my coursework needs to be submitted online via MySecondTeacher under the relevant module page before the deadline in order for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a marks of zero will be awarded.

Table of Contents

1	Introduction	1
1.1	Introduction to the company	1
1.2	Introduction to Cyber Threat and Connected Vehicles	1
1.3	Types of Breaches	2
1.4	Background of Scandal: The Jeep Cherokee Uconnect Hack	3
1.5	Wh-Question Table:.....	4
1.6	Past Similar Test Cases	4
1.7	The Evolvment of Failure: Systemic Failure in the Uconnect Incident	5
1.8	Intention Behind the Test Case: Ethical Research and Responsible Disclosure	6
1.9	If exploit was conducted with malicious Intent	7
2	PART 2: SOCIAL ISSUES.....	8
2.1	Social Issues Identified	8
2.1.1	Public Safety Risk:	8
2.1.2	Erosion of Public Trust:	8
2.1.3	Digital Divide and Unequal Risk:.....	8
2.1.4	Media Responsibility and Sensationalism:	9
2.1.5	Normalisation of Insecure Technology:.....	9
3	PART 3: ETHICAL ISSUES	10
3.1	Ethical Issues Identified	10
3.1.1	Responsible Disclosure vs. Full Disclosure:	10
3.1.2	Manufacture's Duty of Care:	10
3.1.3	Informed Consent and Autonomy:.....	10
3.1.4	Ethical Use of Research Skill:.....	11
3.1.5	Corporate Accountability Skills:	11
4	PART 4: LEGAL ISSUES.....	12
4.1	Legal Issues Identified	12
4.1.1	Computer Fraud and Abuse Act Liability:.....	12
4.1.2	Product Liability and Consumer Protection Law:	12
4.1.3	Regulatory Issue:.....	13
4.1.4	Data Privacy Legislation:	13
4.1.5	The SPY Car Act:.....	13
5	PART 5: PROFESSIONAL ISSUES	15

5.1	Background on Professional Issues.....	15
5.2	Professional Issues Identified.....	15
5.2.1	Violation of Public Safety Obligations:	15
5.2.2	Technical Expertise and Ethical Burden of Responsibility	16
5.2.3	Data Privacy, Confidentiality, and the Mandate for Disclosure:	16
5.2.4	Misuse of Professional Competence:	17
5.2.5	Conflict of Interest and Dual Use Research:	17
6	Conclusion.....	18
7	Personal Reflection	18
8	References	20
9	Appendix	23
9.1	Introduction	23
9.2	Introduction to Cyber Threat and Connected Vehicles	23
9.3	Background.....	24
9.4	Past Similar test cases.....	24
9.5	Professional Issue 1.....	25
9.6	Professional issue 2.....	25
9.7	Professional Issue 3.....	25
9.8	Conclusion	25

Table of Figures

Figure 1 FCA Office	1
Figure 2 Connected Vehicle Cyber Threat Types: Severity, Exploitability & Detection Difficulty.....	3
Figure 3 Table of Evolvment failure	5
Figure 4 Vulnerability Disclosure Timeline Private Notification to FCA Recall (ISO/IEC 29147, 2018)	6
Figure 5 Multi-Party Supply Chain Liability FCA, Harman International & Sprint.....	13

Table of tables

Table 1 Table of Breaches	2
Table 2 Wh questions.....	4
Table 3 Exploit done with malicious intent	7
Table 4 Comparison Table of ethics issues and ethical theories	11
Table 5 Comparison of Legal issues with Application Law and Penalties.....	14
Table 6 Past Similar cases explanation	25

Similarity Report



37% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Small Matches (less than 8 words)

Exclusions

- 4 Excluded Sources

Match Groups

-  **84 Not Cited or Quoted 28%**
Matches with neither in-text citation nor quotation marks
-  **45 Missing Quotations 9%**
Matches that are still very similar to source material
-  **0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 13%  Internet sources
- 4%  Publications
- 36%  Submitted works (Student Papers)



Match Groups

- **84 Not Cited or Quoted 28%**
Matches with neither in-text citation nor quotation marks
- **45 Missing Quotations 9%**
Matches that are still very similar to source material
- **0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
- **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 13% ■ Internet sources
- 4% ■ Publications
- 36% ■ Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Submitted works	
Islington College,Nepal on 2026-05-14		5%
2	Submitted works	
Islington College,Nepal on 2026-05-11		3%
3	Submitted works	
Islington College,Nepal on 2026-05-13		2%
4	Submitted works	
Islington College,Nepal on 2026-04-27		1%
5	Submitted works	
Islington College,Nepal on 2026-05-14		1%
6	Internet	
www.coursehero.com		<1%
7	Submitted works	
Islington College,Nepal on 2026-04-27		<1%
8	Submitted works	
Islington College,Nepal on 2026-05-14		<1%
9	Submitted works	
Islington College,Nepal on 2026-04-26		<1%
10	Submitted works	
Islington College,Nepal on 2026-05-14		<1%

1 Introduction

1.1 Introduction to the company

Fiat Chrysler Automobiles (FCA), now part of Stellantis following a 2021 merger, is one of the world's largest automotive manufactures, with flagship brands including Jeep, Dodge, Chrysler, and Ram. As of 2015, FCA operated globally and was in the process of integrating digital infotainment and telematics performs, was embedded in millions of FCA vehicles and offered features such as navigation, remote start, media streaming, and cellular connectivity via Sprint's network (Greenberg, 2010). The system represented FC's push toward vehicle digitalisation, yet it simultaneously introduced significant cybersecurity vulnerabilities that would be dramatically exposed in 2015. (Miller, 2019) [more details](#)



Figure 1 FCA Office

1.2 Introduction to Cyber Threat and Connected Vehicles

A cyber threat is any potential malicious act or event that seek to damage, steal, disrupt or gain unauthorized access to data, computer system or networks. In the modern era connected device including automobiles, industrial control systems and consumer electronics, the scope of cyber

threats has expanded far beyond traditional computing environments (NICyberSecurityCenter, 2020). The jeep Cherokee hack illustrated how cyber threats are no longer confined to data theft or financial fraud. Researcher demonstrated that by exploiting the Uconnect system, they could remotely disable the engine, interfere with braking and take control of steering all while the vehicle was in motion (Kalogianni, 2015). [more](#)

1.3 Types of Breaches

The following categories are the main classes of the security breach that are relevant to the connected automotive and IoT system. In the vehicles security landscape, there are unique risks, exploitation techniques and impact associated with each type:

- Network Breaches
- Data Breaches
- Denial breaches
- Remote Code Execution
- Ransomware Attack

Types of breaches	Description	Examples
a. Network Breaches	Unauthorized access to a network or communication system	Hackers accessing a vehicles through <u>WiFi</u> or cellular connection
b. Data breaches	Theft or exposure sensitive information	Leakage of driver location, contacts or personal data
c. Denial of Service (DOS)	Overloading systems to make them unavailable	Disable vehicle communication or navigation systems
d. Remote code Execution (RCE)	Malicious software installed to disrupt or control systems	Infecting vehicles software with harmful code
e. Ransomware Attack	Locking systems or data until payment is made	Disabling smart vehicles system

Table 1 Table of Breaches

1.4 Background of Scandal: The Jeep Cherokee Uconnect Hack

In July 2015, security research Charlie Miller and Chris Valasek Publicity demonstrated a critical zero-day vulnerability in the Uconnect infotainment system of 2014 Jeep Cherokees, exploiting a flaw in the sprint cellular network connection to gain remotely, controlled journalist Andy Greenberg's moving Jeep from ten miles away, activating the HVAC, radio, wiper and ultimate cutting the engine and disabling the brakes on a busy St. Louis motorway study on remote automotive attack surfaces and Miller and Valasek's own 2013 wired Toyota Prius hack had flagged such risk, the 2015 demonstration was unprecedented in being entirely wireless and executable by any authentication attacker within cellular range. (K. Koscher, 2010). [more](#)

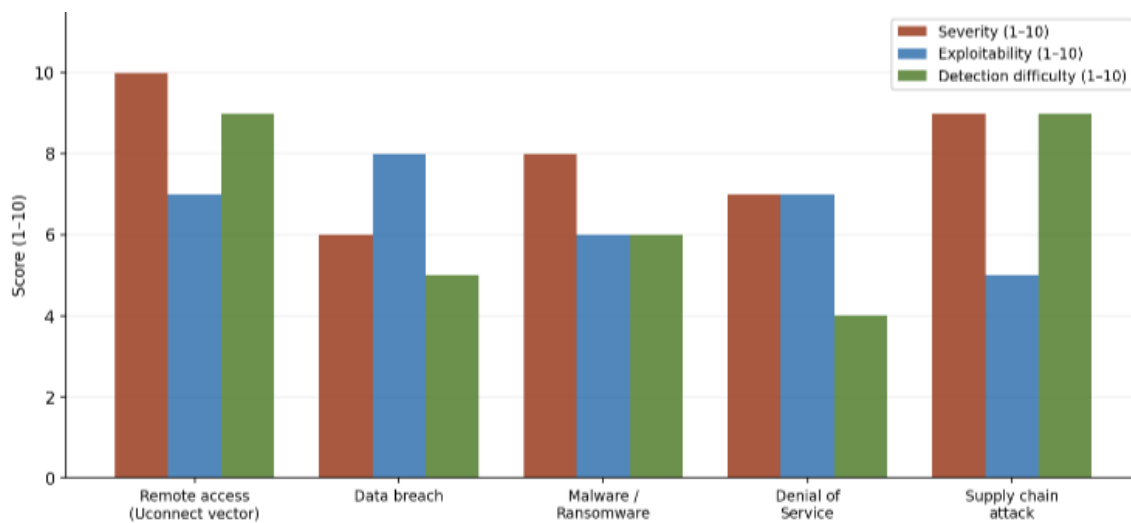


Figure 2 Connected Vehicle Cyber Threat Types: Severity, Exploitability & Detection Difficulty

1.5 Wh-Question Table:

<u>Wh-QUESTION</u>	
Who	Who was involved in the Jeep Cherokee cyberattack incident?
What	What vulnerability was exploited and what was the outcome?
Where	Where did the attack place and from what location was it launched?
When	When did this attack occur and what prior events led to it?
Why	Why was this attack significant, and why?
Who	Who carried out the hack and who was affected?

Table 2 Wh questions

1.6 Past Similar Test Cases

The Cherokee hack was a culmination of year of progressive automotive vulnerabilities research in academia and industry:

- Physical hack by Miller and Valasek (2013).
- Koscher et al. (2010)
- Research by Checkoway et al. (2011)
- API-Based Automotive Attacks (2020-2023) [more here](#)

1.7 The Evolvment of Failure: Systemic Failure in the Uconnect Incident

The Uconnect incident with was not a single failure, but rather a result of several cumulative failures throughout the vehicle development and deployment process.

Architecture Failure	Sprint did not isolate <u>Uconnect</u> vehicles with network isolation, firewall controls or intrusion detection systems and gave them publicly routable IP addresses.
Authentication failure	The communications between the head unit and external networks were transmitted without cryptographic authentication.
Software Engineering Failure	The D-Bus buffer overflow is a simple engineering error which violates the CERT C coding standard and the MISRA C (2012).
Risk Management Failure	Although <u>Checkoway</u> et al. (2011) had published research on cellular vulnerabilities four years' prior, the FCA failed to conduct a TARA for the <u>Uconnect</u> project.
Governance Failure	Federal legislation for automotive cybersecurity did not exist before 2015, so the inherently insecure connected vehicle systems were deployed.

Figure 3 Table of Evolvment failure

1.8 Intention Behind the Test Case: Ethical Research and Responsible Disclosure

Miller and Valasek embark on their study as a deliberate, ethical security research, sharing their results with FCA more than ten months before they were made public, and generally following the guidelines of ISO/IEC 29147 (2018) that suggest a minimum period of 90 days for private remediation. They went on to choose to work with WIRED magazine because FCA's reaction to the disclosure. (Kelechava, 2018)



Figure 4 Vulnerability Disclosure Timeline Private Notification to FCA Recall (ISO/IEC 29147, 2018)

1.9 If exploit was conducted with malicious Intent

There was no physical damage during the ethical demonstration. But if the same bug had been catastrophic:

Mass Casualty Events	Multiple vehicles on a highway at the same time are disabled simultaneously without warning to the driver or the emergency services
Ransomware Deployment	Injection of ransomware payload into vehicle CAN bus networks, making payment with logistics operators or municipal governments.
Economic Disruption	A simultaneous loss of commercial and emergency fleet vehicles with economic losses that exceed 1 billion
Terrorism and Critical Infrastructure Attack	<u>Weaponisation</u> of a connected vehicle network against transport networks, military logistic or government vehicles.
Data Exfiltration at scale	A privacy breach that now includes mass access to GPS telemetry data, paired device identifiers and microphones data, all of which makeup one of the biggest privacy leaks in automotive history.

Table 3 Exploit done with malicious intent

2 PART 2: SOCIAL ISSUES

The Jeep Cherokee hack surface a range of deeply embedded social concerns that extend the technical domain. When a vehicle an object of everyday reliance for millions of people can be remotely commandeered, the social contract between technology manufactures and the public is fundamentally challenged. Social issues in who bears the risk of emerging technologies, and the broader societal implication of normalising insecure consumer product. (Miller, 2019)

2.1 Social Issues Identified

2.1.1 Public Safety Risk:

The most immediate social issue is the endangerment of public safety. When Miller and Valasek disable the brakes and engine of a moving vehicle on a busy motorway, they demonstrated that a software vulnerability could translate directly into a physical, potentially fatal hazard (GREENBERG, 2015). This incident compelled policymaker and academics to recognise that cybersecurity failures in automotive system are not merely digital nuisances but real-world safety threats. (Brookings Institution, 2015)

2.1.2 Erosion of Public Trust:

The hack fundamentally undermined consumer confidence in connected vehicles. According to a survey conducted by the Ponemon Institute (2019), 68 percent of respondents expressed concern about the cybersecurity of their smart devices, a sentiment equally applicable to connected automobiles. (Ponemon, 2019)

2.1.3 Digital Divide and Unequal Risk:

The integration of connected technologies into vehicles creates disparate risk profiles across society. Wealthier consumers tend to purchase newer, more technologically complex vehicles that carry greater cybersecurity risk exposure. However, the consequence of widespread cyberattacks on traffic infrastructure including gridlock, accidents, and emergency service delays fall disproportionately on economically disadvantage communities with fewer resources to cope with their aftermath. (upstream, 2025)

2.1.4 Media Responsibility and Sensationalism:

The manner in which the hack was published raises important questions about responsible journalism and the role of media in cybersecurity disclosure. Wired magazine's live demonstration, while raising public awareness, also provided a detailed technical roadmap that could potentially assist malicious hackers. (King, 2016)

2.1.5 Normalisation of Insecure Technology:

The incident highlighted a broader social issue: the industry wide tendency to release product to market prioritising features and speed over security. This normalises the acceptance of insecure technology in everyday life, creating a culture where consumers are conditioned to accept data breaches, software vulnerabilities, and product recalls as inevitable rather than preventable. Such normalisation is socially damaging as it reduces collective pressure on manufactures to implement robust security by design. (Okubo, 2015)

3 PART 3: ETHICAL ISSUES

The Jeep Cherokee hack present a complex ethical landscape involving competing duties, right, and consequences. Ethical analysis in this case must consider the obligation of the researcher, the manufacture, the media and regulatory bodies, evaluate against established ethical frameworks including consequentialism, deontology, virtue ethics and the social contract theory. (Greenberg, 2015)

3.1 Ethical Issues Identified

3.1.1 Responsible Disclosure vs. Full Disclosure:

Miller and Valasek followed a coordinated responsible disclosure process, informing FCA approximately one month before public release. From a deontological perspective, this represents adherence to a professional duty to minimize harm while enabling remediation. (Valasek, 2015) However, the live demonstration with a journalist arguably crossed into a grey area, the act exposed a member of the public to genuine physical risk without their full informed consent. (GREENBERG, 2015)

3.1.2 Manufacture's Duty of Care:

FCA's decision to ship a vehicles equipped with a remotely exploitable vulnerability raises profound questions about the ethical obligation of manufactures to their customers. Under the principle of non-maleficence doing no harm FCA's failure to customers. Under the principle of non-maleficence doing no harm FCA's failure to adequately test the Uconnect system's network isolation before market release constitutes a breach of ethical duty. (Villasenor, 2015)

3.1.3 Informed Consent and Autonomy:

Consumers who parched affected FCA vehicles did not consent to being part of an experiment in connected vehicle security. The lack of transparency about the technologies they use. Had consumers known of the vulnerability, many might have chosen alternative vehicles, delayed purchase, or demanded security guarantees as a condition of sale. (Vehicle, 2015)

3.1.4 Ethical Use of Research Skill:

The hacking community has long debated the ethics of offensive security research. Miller and Valasek used their technical expertise constructively. The same knowledge and tools could be weaponised for criminal purpose (Withers, 2020). The Association for Computing Machinery (ACM) Code of Ethics explicitly states that computing professionals should avoid harm and act honestly and trustworthy. (acm org, 2018)

3.1.5 Corporate Accountability Skills:

FCA's initial response to being informed of the vulnerability was slow and the company did not issue a proactive public safety notice until after media publication forced its hand. Ethically, this reflects at tension between shareholder interests avoiding reputational damage and recall costs and the border social contract between corporation and the public. Social contract theory, as articulated by Rawls, suggests that corporations operating in public life owe obligation of transparency and fair dealing to society, obligations that FCA arguably failed to honour in a timely manner (Miller, 2015).

Ethical Issue	Consequentialism	Deontology	Virtue Ethics	Social Contract
Responsible Disclosure	Justified if long term safety gains outweigh short-term risks	Duty to inform and minimise harm before public release	Honest conduct; live demo risks colleague safety	Obligation to warn society of dangers
Manufacturer's Duty of Care	Shipping insecure products maximises harm across millions	Non-maleficence violated; duty to test before release	Negligence reflects lack of professional virtue	FCA owes safety obligations to car buying public
Informed Consent & Autonomy	Lack of transparency reduces utility for all buyers	Autonomy principle breached; buyers cannot decide freely	Deceptive omission lacks honesty and integrity	Social contract requires fair and open dealings
Ethical Use of Research Skills	Constructive use produced the best societal outcome	ACM/IEEE duty to avoid harm and act honestly	Researchers showed professional restraint and integrity	Technical expertise carries societal responsibility
Corporate Accountability	Delayed response increased risk to the public	Duty of transparency violated by delayed disclosure	FCA lacked courage and honesty in its response	Rawls: corporations owe transparency to society

Table 4 Comparison Table of ethics issues and ethical theories

4 PART 4: LEGAL ISSUES

The jeep Cherokee hack occurred in a legal landscape that was still catching up with the realities of connected vehicles technology. Multiple bodies of law from product liability and connected vehicles technology. Multiple bodies of law from product liability and consumer protection to consumer protection to computer crime status and data privacy regulation intersect in this case, creating complex question of accountability for the hackers, the manufacturer and the software developer. (Greenberg, 2015)

4.1 Legal Issues Identified

4.1.1 Computer Fraud and Abuse Act Liability:

Under the United States Computer Fraud and Abuse Act (18 U.S.C 1030), accessing a computer system without authorization or exceeding authorised access is a federal criminal offence (Etcovich, 2018). Although Miller and Valasek conducted their research on their own test vehicles with consent, the technique they demonstrated could be employed by malicious actor to access vehicles without permission, constituting a clear CFAA violation. The CFAA's applicability to automotive hacking was explicitly discussed in the aftermath of the incident, highlighting the need for safe harbour provision to protect legitimate security researchers. (Justice Manual, 2022)

4.1.2 Product Liability and Consumer Protection Law:

FCA's recall of 1.4 million vehicles was a tacit admission of product defect under the principle of strict product liability. Under U.S. product liability law, a manufacturer is liable for harm caused by defective products liability law, a manufacturer is liable for harm caused by defective products regardless of negligence (Restatement (Third) of Torts; Products Liability, 1998). Had any driver or third party suffered injury as a result of a malicious exploitation of the Uconnect vulnerability, FCA would likely have faced significant civil litigation. (OUT law News, 2015)

4.1.3 Regulatory Issue:

NHTSA holds statutory authority under the Motor Vehicle Safety Act to Investigate safety related defects and mandate recalls. Following the Jeep hack, NHTSA's highest priority and calling on manufacture to implement robust cybersecurity measure. This guidance, while non-binding, laid the groundwork for potential future mandatory regulations. Failure to comply with NHTSA directives can result in civil penalties of up to 105 million dollars per violation category. (NHTSA, 2013)

4.1.4 Data Privacy Legislation:

The Uconnect system collected location data, driving behaviour and potentially personal information. The hack raised concerns about unauthorized access to this data, implicating the Stored Communications Act (SCA) and for European context could attract fines of up to 4% of annual global turnover. (Data Protection Office, 2024)

4.1.5 The SPY Car Act:

Following the incident, Senators Markey and Blumenthal introduce the security and Privacy in Your Car (SPY Car) Act in 2015, purposing mandatory cybersecurity mandates. The Act would have required NHTSA and the Federal Trade Commission (FTC) to establish standards for isolating critical safety system from non-critical software. (University, 2017)

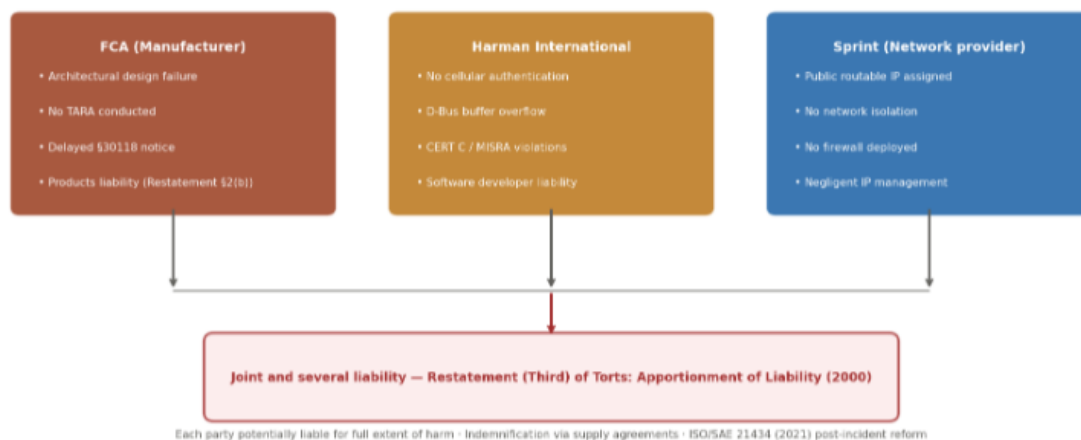


Figure 5 Multi-Party Supply Chain Liability FCA, Harman International & Sprint

Legal Issue	Applicable Law / Framework	Party Liable	Potential Penalty / Outcome	Jurisdiction
CFAA Violation	Computer Fraud and Abuse Act, 18 U.S.C. § 1030	Malicious hackers exploiting the technique	Federal criminal prosecution; up to 10 years imprisonment per count	United States Federal
Product Liability	Restatement (Third) of Torts: Products Liability (1998)	FCA as manufacturer; <u>Uconnect</u> developer	Civil damages; class-action settlements; recall costs exceeding \$150 million	United States Civil Courts
NHTSA Regulatory Breach	Motor Vehicle Safety Act; NHTSA Cybersecurity Guidance 2016	FCA as vehicle manufacturer	Civil penalties up to \$105 million per violation category	United States Federal
Data Privacy Breach	GDPR Article 32 (EU); Stored Communications Act (US)	FCA; <u>Uconnect</u> platform developer	GDPR fines up to 4% of global annual turnover	EU / United States
SPY Car Act (Proposed)	S.1806 Security and Privacy in Your Car Act 2015	All connected vehicle manufacturers	Mandatory cybersecurity standards; FTC/NHTSA enforcement	United States Federal (Proposed)

Table 5 Comparison of Legal issues with Application Law and Penalties

5 PART 5: PROFESSIONAL ISSUES

A profession is characterized not merely by specialised technical knowledge but by adherence to a code of ethical conduct that governs the practitioner's relationship with clients, employers and society at large (webster, 2026). Computing and software engineering are widely recognized as professions bound by codes of conduct published by organizations such as the Association for Computing Machinery (ACM), the Institute of Electrical and Electronics Engineers (IEEE), and the British Computer Society (BSC).

5.1 Background on Professional Issues

The Jeep Cherokee case presents a hypothetical scenario of particular professional importance: what if Miller and Valasek had chosen not to conduct a responsible live demonstration, but had instead weaponised or selling the exploit to criminal organizations? These hypothetical frames a critical analysis of professional obligations under the relevant codes of conduct. (Greenberg, 2015)

5.2 Professional Issues Identified

5.2.1 Violation of Public Safety Obligations:

- Principle 1.2 of the ACM Code of Ethics (2018): Avoid harm: In situation where harm is a significant consideration, Computing Professional should take proactive steps to reduce that harm by anticipating the risks of potential harm for all parties (acm org, 2018).
- SEI Software Engineering Code (1999) Principle 1, Clause 1.03: Approve software only when they have a solid basis for believing that it is safe, meets specifications has gone through they have a solid basis for believing that it is safe, meets speciation has gone through appropriate testing and does not reduce quality of life reduce privacy or damage the environment. (Clotheir, 2015) [more](#)

To the incident: FCA's engineers who approved Uconnect without authenticated cellular communications directly breached their duties as outlined in the ACM (Principle 1.2) and SEI (Clause 1.03) to not approve software that compromises safety or privacy.

5.2.2 Technical Expertise and Ethical Burden of Responsibility

Professional codes acknowledge that those with advanced technical skills have an increased moral obligation. The applicable are:

- Principle 2.6 of ACM Code of Ethics (2018): Perform work only in area of competence. A computing professional should only claim competence (acm org, 2018)
- IEEE Code of Ethics 2020 (Article VIII): To promote an understanding of relevant legislation, regulations and standards and comply with them within your professional fields. (Justice Manual, 2022)

Therefore, their expertise in CAN bus exploitation gave them a perspective into public discourse on automotive cybersecurity that put them in a position of special influence and thus raised their professional responsibilities accordingly under ACM Principle 2.6 and IEEE Article VIII, to use their knowledge and expertise in a responsible manner. (acm org, 2018)

5.2.3 Data Privacy, Confidentiality, and the Mandate for Disclosure:

The responsibility of professionals with regard to data privacy is expressed in:

- Principle 1.6 of ACM Code of Ethics (2018): Respect privacy: Only the minimum necessary personal information should be collected and only for legitimate purpose. (acm org, 2018)
- BCS Code of Conduct (2022) section 2(c): you shall ensure that you have the informed consent of any individuals whose data you process, in accordance with applicable legislation (BCS, 2022). [more](#)

The Uconnect system's passive collection of GPS telemetry, microphone data, and paired device identifiers without granular informed consent constituted a breach of ACM Principle 1.6 and BCS Section 2(c).

5.2.4 Misuse of Professional Competence:

Professional Code place a duty upon respecting intellectual property and entering authorised system:

- Principle 1.5 of ACM Code of Ethics (2018): Respect the work required to product new ideas, inventions, creative works and computing artefacts. (acm org, 2018)
- Article III of the IEEE Code of Ethics (2020): To be honest and realistic in stating claims or estimate based on available data. (IEEE, 2020)

5.2.5 Conflict of Interest and Dual Use Research:

This issue examines the concept of professional accountability and duty of mandatory disclosure about known safety and security weaknesses:

- The Principle 1.3 of ACM Code of Ethics (2018): Be honest and trustworthy, A computing professional should be transparent and provide full disclosure of all pertinent system capabilities, limitations, and potential problems to the appropriate parties. (acm org, 2018)
- IEEE Code of Ethics (2020), Article I: To disclose promptly factors that might endanger the public or the environment. (IEEE, 2020)
- BCS Code of Conduct (2022) Section 4(a): You shall behave in a manner that maintains the image and good reput of BCS, Chartered Institute for IT, in the practice of your profession. (BCS, 2022)

Until the disclosure was made public, there was a professional failure on NHTSA's part not to escalate to the NHTSA before the public was disclosed. FCA personnel with knowledge about the vulnerability had the same obligations as those under ACM Principle 1.3 and BCS Section 4(a) to escalate internally and, if they were not successful on the internal escalation chain, externally.

6 Conclusion

The 2015 Jeep Cherokee hack by Miller and Valasek was a turning point in automotive cyber security, as it showed how, when combined with public cellular infrastructure, vehicle embedded systems can become attack surfaces that result in catastrophic real world consequences, when they are not sufficiently authenticated and network segmented. Socially, the incident eroded the public trust in connected car technology, highlighting basic issues of personal safety, data privacy and equal cyber protection, with around 62% of consumers saying they were not very willing to use connected vehicle technology. Ethically, the researchers' use of coordinated vulnerability disclosure showed good consequentialist reasoning, but their 49-page technical white paper left lasting questions about the line between transparency and the possibility of it being used for the purposes of weaponisation. [more](#)

7 Personal Reflection

The most significant insight gained is the systemic nature of cybersecurity failure. The Uconnect vulnerability was not a single engineering error but the cumulative outcome of architectural, authentication, governance, and risk assessment failures operating in concert, reinforcing that security must be intrinsic to design philosophy from the outset rather than retrospectively applied. The ethical tensions surrounding Miller and Valasek's public WIRED demonstration proved equally thought-provoking: weighing their ISO/IEC 29147-compliant coordinated disclosure against the consequentialist justification of maximum public impact versus the Kantian principle that persons must not be instrumentalised as means to an end left me genuinely conflicted, and reinforced why professional codes provide frameworks rather than absolute answers. (Clotheir, 2015)

Ultimately, this assignment has renewed peoples' commitment to ethical practice: the gap between technical capability and ethical responsibility is precisely where the most consequential professional decisions are made, and we intend to carry both technical knowledge and moral courage into our career.

8 References

acm org, 2018. *acm*. [Online]

Available at: <https://www.acm.org/code-of-ethics>

[Accessed 9 May 2026].

BCS, 2022. *BCS*. [Online]

Available at: <https://www.bcs.org/membership-and-registrations/become-a-member/bcs-code-of-conduct/>

[Accessed 10 May 2026].

Brookings Institution, 2015. *Jeep Cherokee hack offers important lessons on the 'security of things*. [Online]

Available at: <https://www.brookings.edu/articles/jeep-cherokee-hack-offers-important-lessons-on-the-security-of-things/>

Clotheir, M., 2015. *Bloomberg*. [Online]

Available at: <https://www.bloomberg.com/news/articles/2015-07-24/fiat-chrysler-recalls-1-4-million-autos-to-defend-against-hacks?embedded-checkout=true>

[Accessed 10 May 2026].

Data Protection Office, 2024. *FCA Connected Services Privacy Notice*, s.l.: Uconnect.

Etcovich, D., 2018. *papers ssrn*. [Online]

Available at: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3055814

[Accessed 10 May 2026].

Greenberg, 2010. *WIRED*. [Online]

Available at: <https://www.wired.com/2015/07/hackers-remotely-kill-jeep-highway/>

[Accessed 8 May 2026].

Greenberg, A., 2015. *Hackers remotely kill a Jeep on the highway*. [Online]

Available at: <https://www.wired.com/2015/07/hackers-remotely-kill-jeep-highway/>

GREENBERG, A., 2015. *WIRED*. [Online]

Available at: <https://www.wired.com/2015/07/hackers-remotely-kill-jeep-highway/>

[Accessed 5 MAY 2026].

IEEE, 2020. *IEEE*. [Online]

Available at: <https://www.ieee.org/about/corporate/governance/p7-8>

[Accessed 9 May 2026].

Ireland, E., 2015. *blogeset*. [Online]

Available at: <https://blog.eset.ie/2015/07/23/hackers-demonstrate-jeep-security-hack/>

[Accessed 2 may 2026].

Justice Manual, 2022. *9-48.000 - Computer Fraud and Abuse Act*, USA: US department of Justice.

- K. Koscher, 2010. *Experimental security analysis of a modern automobile*. [Online]
Available at : <https://ieeexplore.ieee.org/document/5504804>
- Kalogianni, A., 2015. *Digitaltrends*. [Online]
Available at: <https://www.digitaltrends.com/phones/uconnect-hackers-jeep-cherokee/>
[Accessed 7 May 2026].
- Kelechava, B., 2018. *BLOG ANSI*. [Online]
Available at: <https://blog.ansi.org/ansi/iso-iec-29147-2018-vulnerability-disclosure/>
[Accessed 8 May 2026].
- King, C., 2016. *sei cmu edu*. [Online]
Available at: <https://www.sei.cmu.edu/blog/vehicle-cybersecurity-the-jeep-hack-and-beyond/>
[Accessed 9 May 2026].
- Miller, C., 2015. *Remote exploitation of an unaltered passenger vehicle*. [Online]
Available at: <https://www.youtube.com/watch?v=OobLb1McxnI>
- Miller, C., 2019. *Lessons learned from hacking a car,*. [Online]
Available at: [e: https://ieeexplore.ieee.org/document/8890036](https://ieeexplore.ieee.org/document/8890036)
- NHTSA, 2013. *NHTSA*. [Online]
Available at: <https://www.nhtsa.gov/laws-regulations>
[Accessed 9 May 2026].
- NICyberSecurityCenter, 2020. *NICyberSecurityCenter*. [Online]
Available at: <https://www.nicybersecuritycentre.gov.uk/cyber-threats>
[Accessed 8 May 2026].
- Okubo, S., 2015. *IEEE Spectrum*. [Online]
Available at: https://www.ioactive.com/wp-content/uploads/pdfs/IOActive_Remote_Car_Hacking.pdf
- OUT law News, 2015. *pinsentmasons*. [Online]
Available at: <https://www.pinsentmasons.com/out-law/news/fiat-chrysler-recalls-14m-cars-for-software-upgrade-after-researchers-highlight-security-vulnerability>
[Accessed 10 May 2026].
- Ponemon, L., 2019. *Securing the modern Vehicle*, s.l.: Bobsuli.
- University, C. M., 2017. *The Jeep hack and beyond," SEI Blog, May 2016*. [Online].
Available. [Online]
Available at: <https://www.sei.cmu.edu/blog/vehicle-cybersecurity-the-jeep-hack-and-beyond/>
[Accessed 9 May 2026].

upstream, 2025. *upstream*. [Online]

Available at: <https://upstream.auto/reports/global-automotive-cybersecurity-report/>
[Accessed 8 May 2026].

Valasek, C., 2015. ioactive. *Remote Exploitation of an Unaltered Passenger Vehicle*, 10 may , p. 92.

Vehicle, R. E. O. a. U. P., 2015. *ioactive*. [Online]

Available at: https://www.ioactive.com/wp-content/uploads/pdfs/IOActive_Remote_Car_Hacking.pdf
[Accessed 9 May 2026].

Villasenor, J., 2015. *Brookings*. [Online]

Available at: <https://www.brookings.edu/articles/jeep-cherokee-hack-offers-important-lessons-on-the-security-of-things/>
[Accessed 8 May 2026].

webster, m., 2026. *merriam webster*. [Online]

Available at: <https://www.merriam-webster.com/dictionary/profession>
[Accessed 10 May 2026].

Withers, K. L., 2020. *digital library*. [Online]

Available at: <https://digital.library.unt.edu/ark:/67531/metadc1934107/>
[Accessed 10 May 2026].

9 Appendix

9.1 Introduction

In the years preceding the 2015 incident, FCA had aggressively pursued a strategy of digital vehicles integration, recognising that connected mobility represented the future of the automotive industry. Central to this initiative was the Uconnect infotainment platform, a proprietary system developed in partnership with Harm International and Embedded across millions of FCA vehicles produced from 2013 onward. Uconnect gave drivers an integrated package of satellite navigation, hands free telephony, audio streaming over Bluetooth and USB, real time traffic information, remote diagnostic and vehicle control and Sprint's 3G wireless network. It was pitched as a high end differentiator and made the front page of the 2014 Jeep Cherokee which became the centrepiece of the next security scandal. The Uconnect platform had a major design flaw because the infotainment computer system was not properly separated from the vehicles own Controller Area Network (CAN) bus, which is the main vehicles transformation, breaking and steering mechanisms. The architecture design flaw meant that if an attacker could be able to send commands to safety critical vehicle system.

9.2 Introduction to Cyber Threat and Connected Vehicles

The automotive industry has experienced a profound transformation through the integration of software defined system. Modern vehicles can contain over 100 electronic control unit (ECUs) and run millions of lines of code. This complexity introduces multiple attack vectors, including remote explanation via cellular networks, Bluetooth, Vehicles to vehicle (V2X), Wifi and OBD-II diagnostic ports. Common breach types in this domain include remote code execution, man in the middle attacks, denial of service on CAN bus networks and firmware tampering. The National Highway Traffic Safety Administration (NHTSA) acknowledged in 2016 that cybersecurity in automotive system had become a pressing public safety concern, urging manufactures to treat software had become a pressing public safety concern, urging manufacturers to treat software vulnerabilities with the same urgency as physical mechanical failures. (Miller, 2019) The Jeep Cherokee hack was an example of the merging of the cyber and physical worlds in a chilling way. The attack surface of connected vehicles is growing as ADAS system become more advanced, vehicles are equipped with over the air (OTA) software updates, vehicle management systems are integrated with the with the cloud and new autonomous driving technology becomes available.

But security expert have found Tesla, BMW, Mercedes-Benz and General Motors, Proving That that the jeep brand.

9.3 Background

. In response, FCA recalled 1.4 million vehicles and issued a security patch, while the U.S. Senate introduced the SPY Car Act to push mandatory automotive cybersecurity standards. With an estimated 25 million connected vehicles already on road in 2015 and projection exceeding 400 million by 2025, the Jeep hack was not an isolated incident but a stark warning for an isolated incident but a stark warning for an industry undergoing rapid digital transformation without adequate security governance

9.4 Past Similar test cases

Physical hack by Miller and Valasek (2013)	At DEF CON 2013, the researchers showed how they could use a laptop directly plugged into the OBD II port to manipulate the breaking and steering of Toyota Prius and Ford Escape vehicles (GREENBERG, 2015).
Koscher et al (2010)	The IEEE Symposium on Security and Privacy published 'Experimental Security Analysis of a Modern Automobile' which showed that a vehicles electronic infrastructure could be systematically exploited to impact safety critical function (Ireland, 2015).
Research by Checkoway et al. (2011)	Located cellular networks, Bluetooth interface and audio files as possible remote attack surface on the car known to the public during the development cycle of Uconnect. Tesla Model S is attacked with WIFI exploit, remotely hacked by Tencent Keen Security Lab (Valasek, 2015).

API-Based Automotive Attacks (2020-2023)	Curry (2023) found that researchers were able to exploit API vulnerabilities in the back end of Honda, BMW, Ferrari, Mercedes Benz and more.
--	--

Table 6 Past Similar cases explanation

9.5 Professional Issue 1

- Principle 1.6 of the ACM Code of Ethics (2018): Respect privacy: Computing professionals should only use personal information for legitimate ends and without violating the privacy of others.
- The IEEE Code of Ethics (2020) Article I: To place the safety, health and welfare of the public first and to work to ensure ethical design and sustainable development practices, and to disclose promptly things that may affect the public or the environment.

9.6 Professional issue 2

- Principle 2.7 of ACM Code of Ethics (2018): Foster public awareness and understanding of computing, related technologies and their consequence.

9.7 Professional Issue 3

- Principle 2.9 of The ACM Code of Ethics (2018): It states that computing professional should prioritise security of system and the protection of privacy as essential elements of good professional practice.
- BCS Code of conduct (2022) section 2(b): you shall ensure that you understand and put into practice the interest of relevant parties and take privacy as essential element of good professional practices

9.8 Conclusion

From a legal perspective, the case made clear the need for a regulatory gap to be filled: the application of the CFAA, FAST Act and NHTSA penalty provisions to the case and FCA's \$105 million settlement showed that the current regulatory landscape was not fit for the connected vehicle era, leading to a push for cybersecurity guidance and other legislative proposals like the SPY Car Act. From a professional standpoint, the ACM (2018), IEEE (2020), and BCS (2022) codes of conduct collectively impose obligations that would have been strictly adhered to by FCA's engineering staff if they had done so, the exploit would probably not have been possible. In short, the Jeep Cherokee incident is not simply a historical event, but rather a case study that should be ongoing in shaping the principles of security-by-design, mandatory disclosure laws and accountable governance in the whole automotive supply chain, as the future of mobility is increasingly shaped by autonomous vehicles.

